

Max Li

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science (Coop), GPA: 3.89, CAV: 89

Expected Graduation: Spring 2028

- Relevant Coursework: Object Oriented Programming, Functional Programming

EXPERIENCE

Software Engineer Intern (HCI R&D)

January 2026 – Present

Huawei Technologies Canada

Markham, ON

- Engineered data and validation pipelines using **Python**, **Pandas** for feature discovery and edge-case testing for **LightGBM**-based cross-device gesture detection with strict on-device memory and latency budgets
- Developed model features and diagnostic visualization tools to reduce model failures, increased recall from **97%** to **99%** and reduced false positives by **52%** for commercialization across **millions of devices** in Feb
- Built a low-latency multimodal tutorial agent using **Python**, **asyncio**, **multiprocessing**, and **WebSockets** to stream audio/video and generate real-time feedback via open-source LLMs.

Software Engineer Intern (HCI R&D)

September 2025 – December 2025

Huawei Technologies Canada

Markham, ON

- Engineered multimodal LLM workflows for mobile proactive agents, implementing concurrent prompt processing to minimize user-perceived latency.
- Built a real-time proactive mobile audio agent with **RAG** in **Kotlin** and **Python**, fetching real-time data to enable low-latency client-facing workflows
- Optimized a client-facing map recognition and reconstruction service in **Python** using **Flask**, **OpenCV** through parallelization and prompt engineering to reduce latency by **31%**, built **React** frontend for real-time visualization
- Built Android communication infrastructure in **Java/Kotlin** integrating HTTP and WebSockets for maintainable backend integration, reducing duplicate code and accelerating prototyping speed **3x**

PROJECTS

Notification-service | *Python, RabbitMQ, Docker, Kubernetes, FastAPI*

2026

- Developed a distributed notification service using FastAPI, deploying on local **Kubernetes** cluster (**Minikube**)
- Implemented **RabbitMQ** for backpressure and persistence, **KEDA** for event-driven scaling to **1,000+** msgs/sec
- Configured environment settings through ConfigMaps and Secrets, ensuring application code remained portable

Aire, Flight Scraper | *Python, Playwright, Celery, Redis, RabbitMQ, Pydantic, PostgreSQL*

2025

- Engineered a scalable flight scraper with **React**, **FastAPI** and **Playwright** to scrape airline sources concurrently
- Implemented **Redis** rate-limiting, bypassing anti-scraping measures for reliable data collection
- Queued jobs via **RabbitMQ** for decoupled asynchronous execution, deploying on **GCE** and **Vercel**

Cartly, Web Scraping API | *Java, SpringBoot, Selenium, Go, Gin, Github Actions*

2025

- Built a web-scraping API orchestrator in **Java**, efficiently scheduling requests to ensure high availability of service
- Developed scheduled scraping and notification services in **Go**, supporting async updates and real-time alerts

OTHER EXPERIENCE

Algorithms and Data Structures Instructor

2021 – 2024

Glenforest Computer Science Club

Mississauga, ON

- Competed in inter-school and national programming contests, solving complex algorithmic problems in Python
- Trained **20+** members through weekly Python lectures and group projects

TECHNICAL SKILLS

Programming Languages: Python, Java, Kotlin, TypeScript, Go, C, C++, Bash, HTML/CSS, SQL

Technologies: Spring, SpringBoot, FastAPI, React, Node.js, REST, gRPC, Git, VS Code, Docker, Docker Compose, Kubernetes, AWS, GCP, Github Actions, MongoDB, PostgreSQL, Pandas, NumPy, BeautifulSoup, Selenium, Pytorch, Pydantic, Playwright, MySQL, LightGBM, ArkTS, Kotlin, Jetpack Compose, Android Studio